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DAY 1 AGENDA

- ▶ What is Ansible
- ▶ Why Ansible
- ▶ SSH overview
- ▶ Ansible & SSH
- ▶ Installing Ansible & preparing SSH
- ▶ Ad-hoc commands
- ▶ Inventory file
- ▶ Ansible.cfg file
- ▶ Ad-hoc commands escalation
- ▶ Ansible playbook
- ▶ Ansible modules

WHAT IS ANSIBLE ?

- ❖ [Ansible](#) is a software tool that provides simple but powerful automation for cross-platform computer support.
- ❖ It is used for updates on workstations and servers, cloud provisioning, configuration management, and nearly anything a systems administrator does on a daily basis.



WHY ANSIBLE ?

Command module not idempotent

- ❖ **Idempotent**: An operation is idempotent if the result of performing it once is exactly the same as the result of performing it repeatedly without any intervening actions.
- ❖ **Agentless**: Other tools like (Puppet & Chef) require an agent to be installed on the target device. Ansible only requires an SSH connection to the target device.
- ❖ **Open-source**: Ansible is an open-source community project sponsored by Red Hat.

SSH OVERVIEW

- ❖ OpenSSH is the premier connectivity tool for remote login with the SSH protocol.
- ❖ It encrypts all traffic to eliminate connection hijacking and other attacks.



PC 2

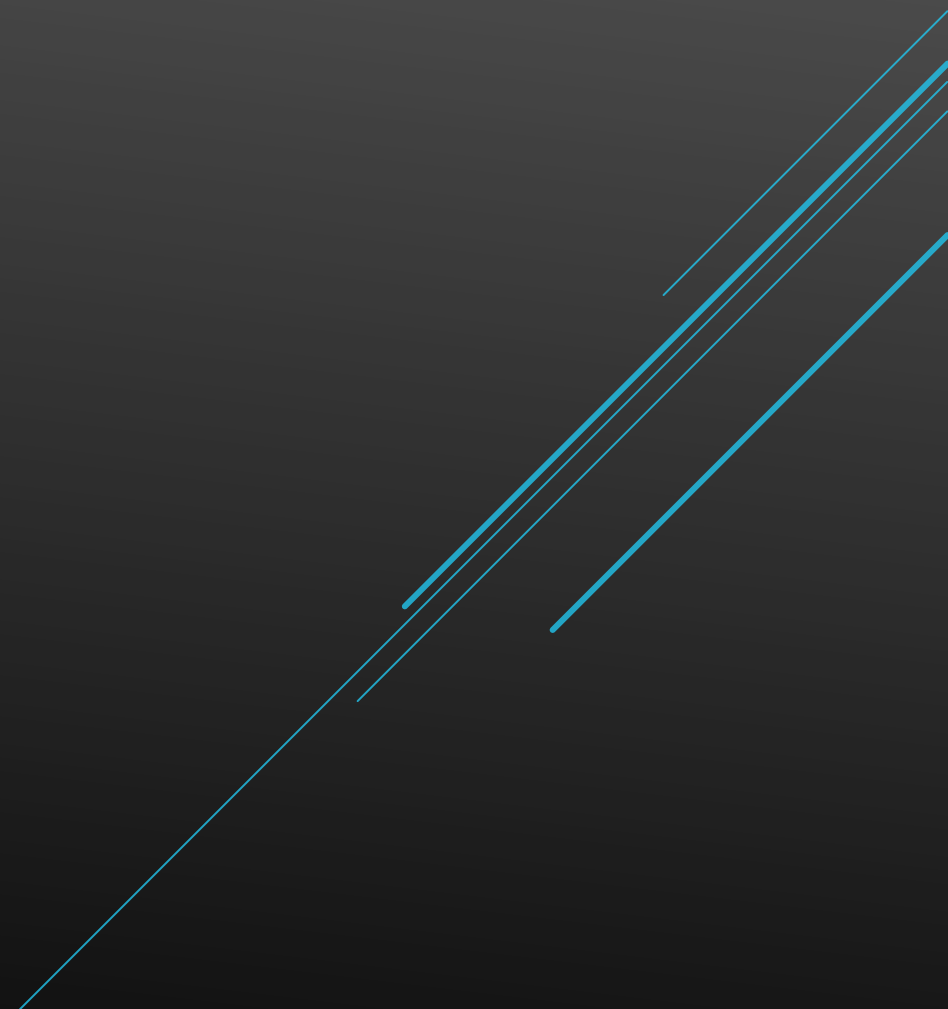


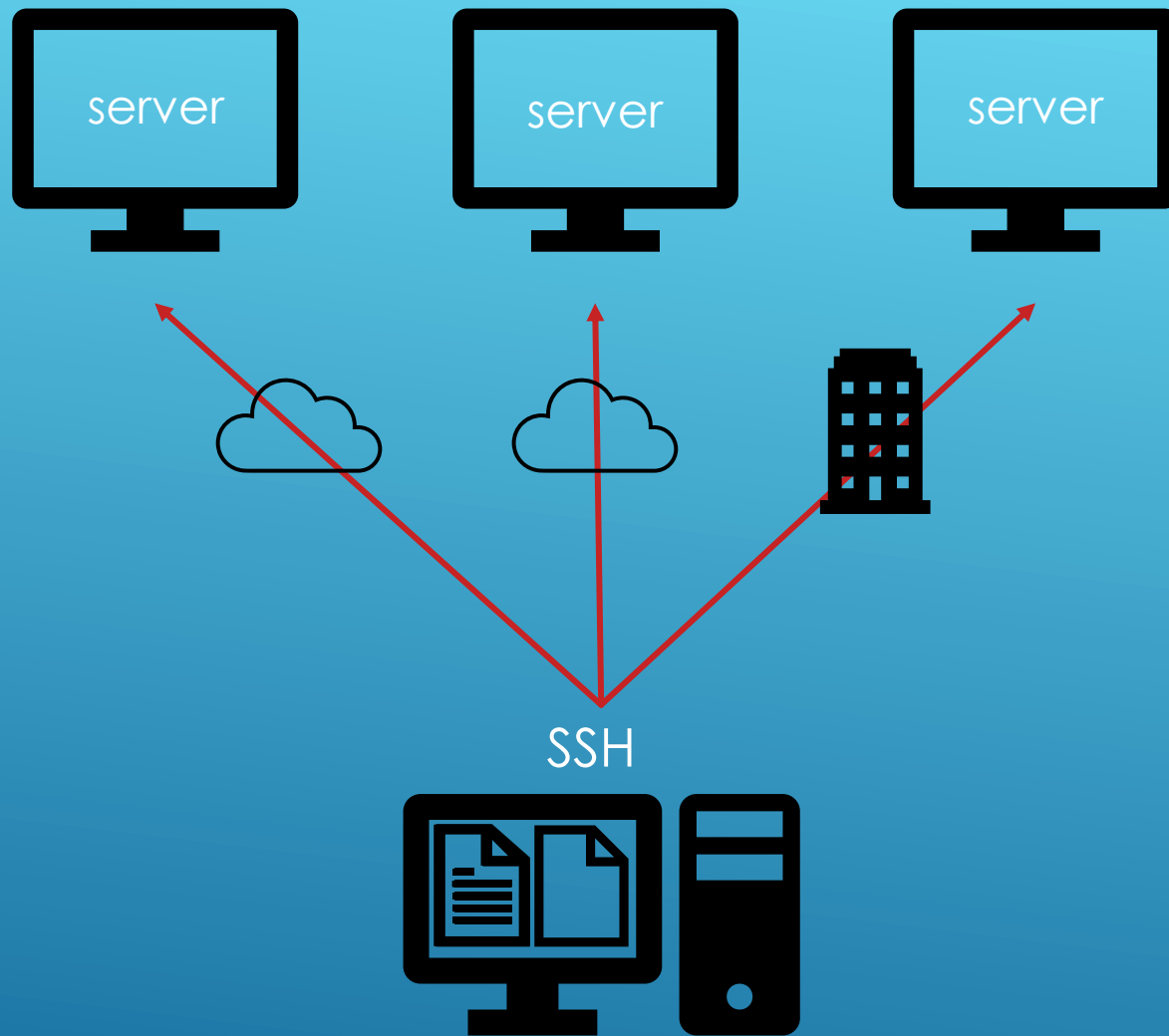
PC 1



ANSIBLE & SSH

How Ansible connects to servers ?



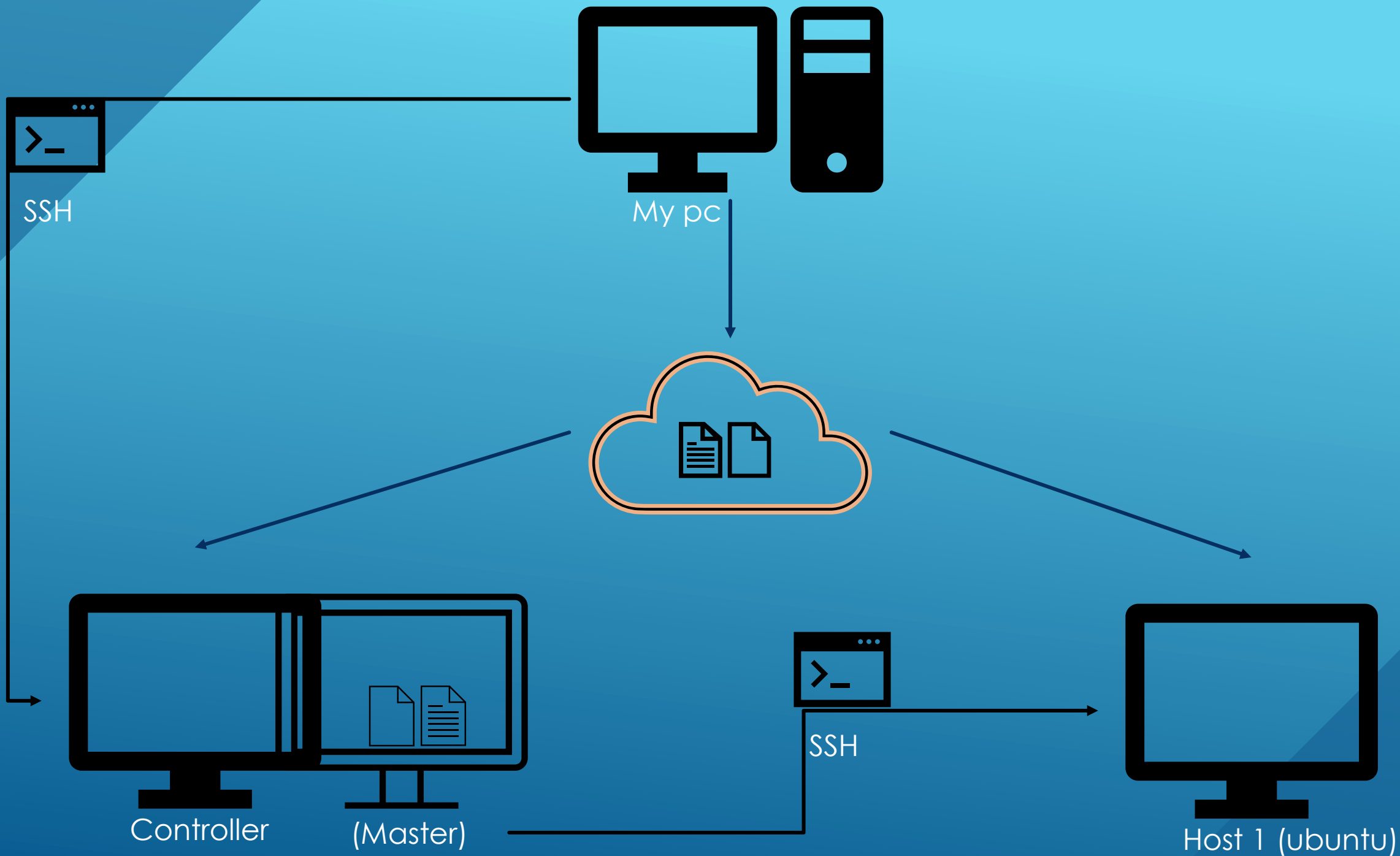


Ansible control machine

- ▶ Install ansible
- ▶ Create a new user on control machine
- ▶ Generate SSH key pair on control machine
- ▶ Copy the public key to host 1
- ▶ Make sure you can ssh into host 1



INSTALLING ANSIBLE & PREPARING SSH



AD-HOC COMMANDS

Ad-hoc: Running Ansible to perform some quick command.

An example of an ad hoc command might be rebooting 50 machines in your infrastructure.

Structure:

```
ansible [pattern] -i [inventory] --private-key [/path/to/private/key] -u [remote_user] -m [module_name]
```

Example:

```
ansible all -i 3.87.24.251, --private-key ~/.ssh/devops -u ubuntu -m ping
```

SSH equivalent:

```
ssh ubuntu@3.87.24.251 -i ~/.ssh/devops -o 'RemoteCommand echo hi; bash;' -t
```

INVENTORY FILE

inventory file: A file that describes Hosts and Groups in Ansible.

Examples:

```
[web_servers]
```

```
3.87.24.251
```

```
[database_servers]
```

```
3.87.24.252
```

```
3.87.24.253
```

- ▶ Create the inventory file
- ▶ Put the IP of host 1 in the inventory file
- ▶ Use the inventory file path in your ad-hoc command instead of using the IP hard-coded
- ▶ Example:
`ansible all -i inventory --private-key ~/.ssh/devops -u ubuntu -m ping`



INVENTORY FILE

CONFIGURATION FILE

ansible.cfg file: This is the brain and the heart of Ansible.

The file that governs the behavior of all interactions performed by the control machine.

Locations:

ANSIBLE_CONFIG (environment variable if set)

ansible.cfg (in the current directory)

~/.ansible.cfg (in the home directory)

/etc/ansible/ansible.cfg

Example:

[defaults]

inventory = ./inventory

private_key_file = ~/.ssh/devops

remote_user = ubuntu

- ▶ Create the configuration file
- ▶ Insert some values in the configuration file
- ▶ Run the minimized ad-hoc command
- ▶ Example: `ansible all -m ping`

CONFIGURATION FILE



AD-HOC COMMANDS ESCALATION

Ad-hoc: Running Ansible to perform some quick command with SUDO permissions.

Structure:

`ansible [pattern] -m [module_name] --become`

Example `ansible.cfg`:

Example:

`ansible all -m command -a "whoami" --become`

`[privilege_escalation]`

`become = true`

- ▶ Insert the correct values in the configuration file
- ▶ Example: `ansible all -m command -a "whoami"`
- ▶ What is the output of the command ?

AD-HOC COMMAND ESCALATION USING ROOT USER



PLAYBOOK

- A **playbook** is a list of **plays**.
- **playbook**: Playbook is the language Ansible uses to orchestrate, configure, administer, or deploy systems.
- **Play**: is a mapping between a set of hosts (groups, hostnames, or IPs) and the tasks which run on those hosts to define the role that those systems will perform.
- There can be **one or many plays** in a playbook.

Examples:

```
- name: your play name
  hosts: all
  tasks:
    - name: your task name
      ping:
```

- ▶ Write your first playbook file
- ▶ Stop gather_facts and update cache

PLAYBOOK



MODULES

Modules: are units of code that can control system resources or execute system commands. Ansible provides a module library that you can execute directly on remote hosts or through playbooks.

Playbook example:

```
- name: your play name
  hosts: all
  tasks:
    - name: your task name
      ping:
```

Ad-hoc command example:

```
ansible all -m ping
```

- ▶ Explore some built-in modules like:

(**apt, dnf, package, service, command, copy, user, group, lineinfile, authorized_key, etc.**)

[ansible-builtin modules](#)

- ▶ Update cache
- ▶ Install latest nginx
- ▶ Copy index.html from controller to host 1
- ▶ Restart nginx service
- ▶ Can you see your index.html file when you hit host 1 on port 80 ?



MODULES

QUESTIONS ?

THANK YOU